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Short Communication

Data welfare is animal welfare: Building a WellFAIR research ecosystem

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1. Introduction

In early September, experts gathered in Helsinki for the closure of the COST Action TEATIME initiative (CA20135), a four-year effort to describe animal behaviours with minimal impact by monitoring them directly in the home-cages where they live. The consortium's outputs, including a comprehensive systematic review (Kahnau et al., 2023) and the definition of essential metadata standards (Moresis et al., 2024), demonstrated that harmonised protocols can drastically reduce inter-laboratory variability. This harmonisation allows researchers to detect subtle behavioural phenotypes with fewer animals, directly supporting the 3Rs.

Yet as the meeting concluded, a familiar paradox emerged: although conceptual consensus was strong, the infrastructure required to sustain rigorous, interoperable behavioural research remains fragile. This disconnection illustrates a broader systemic problem in preclinical research. The principles of Findability, Accessibility, Interoperability, and Reusability (FAIR) (Wilkinson et al., 2016) have effectively joined the 3Rs (Replacement, Reduction, and Refinement) as policy mandates, but the reality of laboratory data production remains far removed from these ideals. Because data is the only enduring outcome of animal experimentation, its loss or degradation represents not just a scientific failure, but an ethical one.

2. The FAIR mandate gap

The magnitude of this "FAIR mandate gap" is alarming. Recent quantitative assessments paint a sobering picture of the field: Tavares et al. (2023) estimate that approximately 80% of preclinical imaging experiments fail to reproduce across sites (Tavares et al., 2023), while a major 2025 report by de Oliveira Andrade highlights that only about one in five protocols from a national replication initiative could be independently reproduced (De Oliveira Andrade, 2025). Similarly, Errington et al. (2021) demonstrated that less than half of attempted cancer biology experiments could be successfully replicated (Errington et al., 2021). The economic implications are staggering—estimated at $28 billion annually in the US alone (Freedman et al., 2015)—but the scientific consequences are even greater: we are attempting to build AI models on a foundation of fragmented, often poorly contextualised data.

3. A layered approach to standardisation

To bridge this gap, we must first diagnose the standards landscape. Global bodies like the Research Data Alliance (RDA) and ELIXIR define overarching interoperability principles, but "good data" cannot be defined by a single universal standard. A "one-size-fits-all" schema is impossible because the scientific requirements of a neuroscientist differ fundamentally from those of a pharmacologist. What is needed is a layered approach to standardisation.

The foundation must be a Minimum Metadata Set (MNMS). Championed by working groups from TEATIME and published in Nature's journal *Lab Animal* (Moresis et al., 2024), the MNMS defines the absolute minimum descriptors—genotype, sex, age, housing conditions, timestamps—that must accompany any dataset to ensure basic

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interpretability. But above this minimal layer, each research community requires its own *Deep Metadata* schema to support domain-specific analysis. The International Mouse Phenotyping Consortium (IMPC) (Groza et al., 2023) proved that strict, centralized schemas can create a transformative functional catalogue of mammalian gene function. Similarly, the neurophysiology community has converged on the Neurodata Without Borders (NWB) data standard (Rübel et al., 2022), while toxicology has adopted the CDISC-SEND standard (CDISC, 2023). These specialised schemas are not obstacles to integration: they are prerequisites for multi-modal analysis, enabling correlations between behavioural motifs (TEATIME) and neural firing patterns (NWB) across independent studies. While the MNMS ensures baseline interpretability, *Deep Metadata* captures domain-specific experimental logic—such as task parameters, acquisition modalities, or analysis assumptions—required for meaningful reuse within a given scientific community.

4. From retroactive curation to "Born-FAIR" data

Despite the availability of standards, most failures of FAIR implementation arise not from missing rules, but from when and how metadata is captured during the research process. Indeed, even when standards exist, researchers often conflate three distinct layers: what must be described (schemas), how concepts relate (ontologies), and how information is technically exchanged (file formats and packaging). Treating these layers as interchangeable produces paralysis and drives researchers back to the perceived safety of spreadsheets.

Spreadsheets are often criticised as engines of semantic chaos (Ziemann et al., 2016), but recent evidence suggests that they can be rehabilitated if the workflow shifts from retroactive to synchronous recording (Seep et al., 2024). Seep et al. (2024) propose a "Metadatasheet" approach in which structured Excel templates—acting as "Metadata Workbooks"—are initialized at the Planning Stage, not retrospectively at the publication stage (Seep et al., 2024). By mapping fields to ontologies before the first mouse is genotyped, researchers can capture context "live," avoiding the data loss inherent to post-hoc curation.

However, migrating from spreadsheets to Electronic Lab Notebooks (ELNs) does not automatically solve the problem. As Vandendorpe et al. (2024) highlights, digitization does not equal FAIRness (Vandendorpe et al., 2024). Without explicit strategies for interoperability and export, ELNs can easily become proprietary silos—digital dead-ends that trap data in non-standard formats. An ELN must be a node within an ecosystem, not an insulated archive. In that direction the ELN Consortium is a collaborative effort to define and adopt open standards, which ensures seamless data exchange between different ELN systems and empowers institutions and researchers to build reproducible, transparent, and sustainable digital research infrastructures.

The critical distinction is therefore not between “manual” and “digital” tools, but between workflows that attempt to repair context after the experiment and those that preserve it from the outset. A viable solution therefore combines Seep et al.’s "Planning Stage" rigor with genuine **"Born-FAIR" (or FAIR-by-design) middleware**. Unlike retroactive "FAIRification"—which attempts to clean up data after the experiment is finished—Born-FAIR tools capture metadata at the source. Tools like Madbot and Metadatapp (Denecker and Seiler, 2024; Huzard, 2025a, 2025b) exemplify this category. Sitting between the bench, the ELN, and the repository, these systems enforce metadata schemas that align terminology with ontologies, and generate machine-actionable outputs (e.g., RO-Crates or JSON-LD) during acquisition. For example, several emerging middleware platforms implement this approach by providing domain-agnostic, schema-driven layers that link experimental design, animal-level metadata, and procedural context across the study lifecycle (e.g., Madbot, Metadatapp). These tools enable FAIR-by-design capture rather than post-hoc FAIRification. This ensures that the digital record is not merely a narrative PDF, but a structured, reusable, machine-actionable and interoperable dataset.

5. Shadow data and the AI challenge

This rigorous structuring is not only about reproducibility: it is essential to fulfil the ethical mandate of the 3Rs, particularly in the era of Artificial Intelligence (AI). A dataset may be technically FAIR yet scientifically useless for machine learning if it reflects biased sampling or incomplete metadata. This leads directly to the issue of **"Shadow Data"**—negative, null, or ambiguous results that traditionally languish in laboratory drawers, remaining unpublished.

In the AI era, these are no longer just "failed experiments”, but critical counterfactual and negative training examples. Models trained exclusively on positive, published results inherit the publication bias of the literature, leading to 'shortcut learning'—where an AI learns to recognize artifacts of successful experiments rather than genuine biological signals—resulting in poor generalizability when applied to new biological contexts (Kapoor and Narayanan, 2023). The FAIR^2 initiative (2025) argues that "AI-Readiness" requires governance structures to surface, validate, and contextualise this shadow data (FAIR^2 Alliance, 2025). Crucially, FAIR compliance is not a proxy for scientific validity. A dataset can be perfectly FAIR yet statistically underpowered (Wilkinson et al., 2016); FAIR simply ensures that such flaws are transparent rather than hidden.

Improved (meta)data management directly strengthens the principle of Reduction. The eTRANSAFE initiative and the emerging VICT3R consortium illustrate the promise of "Virtual Control Groups" (Pognan et al., 2021). By aggregating harmonised historical control datasets from industry into FAIR-compliant repositories, researchers could theoretically replace concurrent control groups with virtual cohorts. Early analyses suggest a potential reduction of animal use by up to 25%—but only if large volumes of standardised, interoperable historical data exist.

6. Federated infrastructure: sovereignty with interoperability

In practical terms, a federated infrastructure separates where data is stored from how it is discovered and accessed, allowing institutions to retain control while still participating in a shared research ecosystem. Finally, we must address the most persistent barrier to data sharing: the fear of losing control. Institutions are often reluctant to upload proprietary datasets in centralised repositories due to concerns around IP, GDPR, and competitive advantage. The solution is to shift from a "Centralized" to a **"Federated"** model. The open-source FAIR3R initiative (FAIR3R Initiative, 2025) embodies this shift. Importantly, federation does not prescribe a single technical solution, but a governance model in which interoperability is achieved without transferring ownership or custody of sensitive data.

FAIR3R addresses the sovereignty deadlock by ensuring that **data stays local, while metadata goes global**. Unlike central repositories that demand custody, FAIR3R acts as a configurable ecosystem. The size of the instance is defined by the stakeholder—it can be deployed at the level of a single research group, a pharmaceutical company, or a national body like the CNRS. Beyond simple hosting, a FAIR3R instance acts as a centralizing Hub, linking local *in vivo* data with related external datasets (such as omics) to unify the scientific narrative. It supports the full research lifecycle, enabling private collaborative work before public release. Upon publication, global discoverability is secured through the automated generation of DataCite DOIs derived from the MNMS-compliant metadata. By decoupling custody from discovery, FAIR3R resolves the tension between institutional sovereignty and global interoperability. In this model, federation is not a compromise, but a prerequisite for scaling FAIR practices across heterogeneous institutional and regulatory contexts.

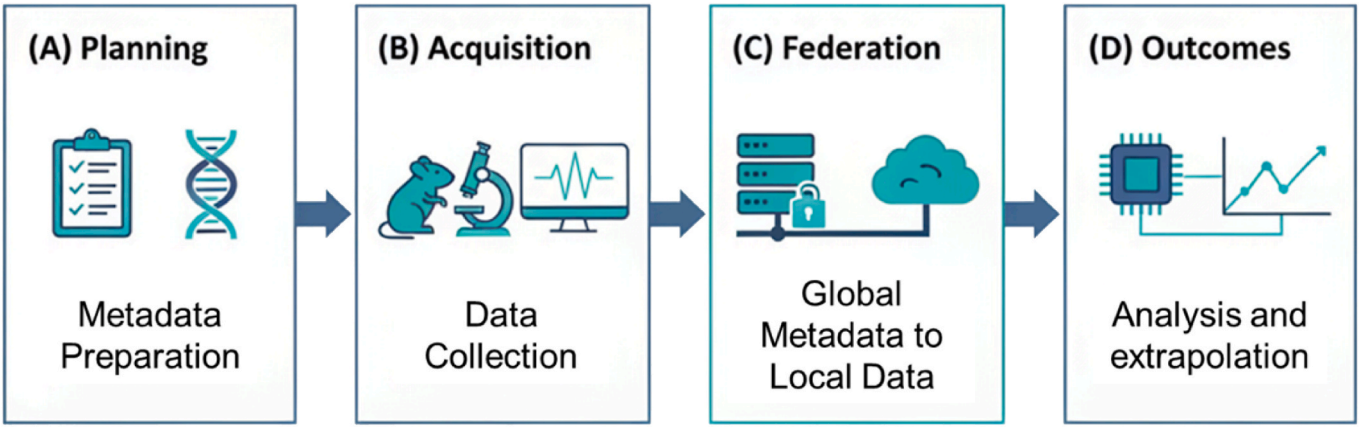

Fig. 1. The WellFAIR Ecosystem.
A schematic representation of the data lifecycle designed to support the 3Rs. (A) Planning: Researchers engage with Data Stewards to select a Minimum Metadata Set (MNMS) before the experiment begins (B) Acquisition: "Born-FAIR" middleware (e.g., Madbot, Metadatapp). acts as a bridge between instruments and Electronic Lab Notebooks (ELNs), capturing metadata in real-time. (C) Federation: The FAIR3R model resolves sovereignty concerns by keeping data on local institutional servers while publishing searchable metadata globally. (D) Outcomes: This structured pipeline enables the aggregation of "Shadow Data" to reduce AI bias and creates sufficient historical datasets to form Virtual Control Groups, directly reducing animal usage.

7. Bridging the sociological gap

Technology alone, however, cannot solve a cultural problem. The expectation that bench scientists should simultaneously become experts in ontologies is unrealistic. We must institutionalize the Data Steward—not as a "Digital Janitor" hired to clean up messy Excel sheets at the end of a grant, but as an Architect. Emerging models like the UK's BioFAIR (BioFAIR UK, 2024) (creating a dedicated national infrastructure) and France's Recherche Data Gouv (Recherche Data and Gouv, 2024) (providing distributed support via Data Workshops) demonstrate that professional support is essential. These stewards are the bridge builders who map local terminology to global ontologies, configure the FAIR-enabling tools at the planning stage, and implement the MNMS within the laboratory workflow.

Finally, we must bridge the sociological gap that frequently separates the "3R/Welfare" community—veterinarians and ethics bodies—from bench scientists. This separation creates a disconnect where ethical oversight and daily data production operate in parallel silos. The **"Culture of Care"**—traditionally defined as a commitment to animal welfare beyond legal minimums—must **naturally extend** to include "Data Care." We must reimagine regulatory training to encompass not just the physical well-being of the animal, but the integrity of the data it produces. Experimenters must be trained to recognize that data is the definitive outcome of the animal's use; therefore, preserving that digital legacy through rigorous management is as ethically binding as ensuring daily physical welfare. To achieve this, we need a dual drive: top-down, stakeholders and funding agencies must provide the necessary human resources and incentives for FAIRification; bottom-up, the research community must cultivate a mindset that pushes beyond minimum requirements to treat data quality as an intrinsic part of their ethical responsibility. This shift can be accelerated by valuing “Short Notes” (as created by French FC3R), 'Data Papers' and 'Micro-publications' of negative results as highly as traditional discovery articles.

8. Conclusion

The gap between FAIR mandates and laboratory reality is not just a technical failure; it is an ethical breach. Using animals in research entails a moral contract: their lives are exchanged for knowledge. When that knowledge is lost in a spreadsheet, trapped in a proprietary ELN silo, or buried as a "negative result," that contract is broken. The path forward requires a coordinated ecosystem: Standards (MNMS and community schemas) to define the language; Human Infrastructure (stewards as architects) to implement it; Born-FAIR (or FAIR-by-design) Tooling (Madbot, Metadatapp and Metadata Workbooks) to capture it; Federated infrastructures (FAIR3R and FairDomHub) to share it (see Fig. 1) ; and Enforcement: journals must move beyond passive checklists to active verification, making data quality a criterion for peer review (Hair et al., 2019). Together, these layers also support an emerging ethical dimension of WellFAIR: Digital Sobriety (Lannelongue et al., 2021) —the ability to identify which data is sufficiently described to be reused, archived, or responsibly deleted. We cannot simply hoard exabytes of data. We require rich metadata to identify which data is redundant and can be deleted "FAIR for Deletion".

This is the essence of WellFAIR: only by aligning these structural components can we ensure that each dataset contributes meaningfully to the global evidence base, strengthens the reproducibility of AI-driven science, and honours our responsibility to the animals whose lives make biomedical progress possible.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Damien Huzard reports a relationship with Metadatapp that includes: employment. Dr. Damien Huzard is the founder and scientific lead of Metadatapp, a developing platform focused on structured metadata management and FAIR data practices in preclinical research. This activity is independent of the work reported in this manuscript and did not provide funding, data, or infrastructure for the study Metadatapp is mentioned in the manuscript as an illustrative example of emerging tools in the metadata and FAIR data ecosystem. This mention does not influence the analysis or conclusions of the article. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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